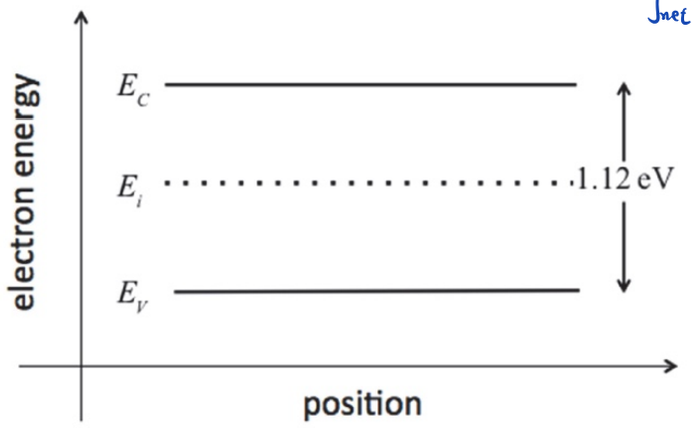
Homework 6 (due on 4/20/2026)

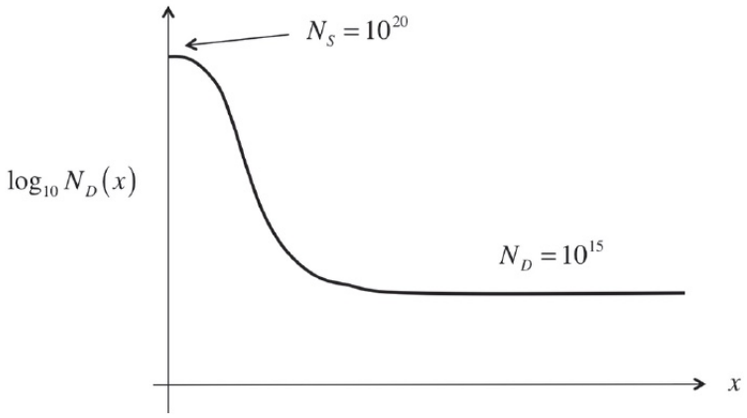
1. Consider silicon at T =300 K that is doped with donor impurity atoms to a concentration of $N_d=1\times10^{17}$ cm^{-3} , the intrinsic carrier concentration is $n_i= 1\times10^{10}$ cm^{-3}, the excess carrier lifetime is 1×10^{-6} s.
   1) If we add excess electrons and holes so that $\delta n=\delta p =10^{12}$ cm^{-3}, is the semiconductor in low-level injection? Explain why or why not. (10pt)
   2) Where are the quasi-Fermi levels, Fn and Fp located with respect to the intrinsic, Ei? Provide the numerical value and show the Fn and Fp on the energy band diagram below. (10pt)



   3) Determine the thermal equilibrium recombination rate of holes. (10pt)
   4) Excess carriers are generated such that $\delta n=\delta p =10^{12}$ cm^{-3}. What is the recombination rate of holes for this condition? (10pt)

2. When computing SRH recombination rates, we usually focus on defects with energy levels near the middle of the bandgap. Beginning with the SRH expression, please show that states near the middle of the bandgap have the largest effect on the SRH recombination rate. (20pt)

3. The doping profile for an n-type silicon wafer ($N_d = 10^{15}$ cm^{-3}) with a heavily doped thin layer at the surface (surface concentration, $N_s = 10^{20}$ cm^{-3}) is sketched below. Answer the following questions.



1) In equilibrium condition, compute the position of the fermi level with respect to the bottom of the conduction band at x=0 and as x-> infinity. (10pt)
2) Using the above information, sketch conduction band Ec(x) vs. x. be sure to include the Fermi level. (10pt)
3) Sketch the electrostatic potential vs. position (10pt)
4) Sketch the electric field vs. position (10pt)

1. (1) $N_d = 10^{17}/cm^3$.

$n_0 = N_d = 10^{17}/cm^3$

$\Rightarrow n_0 >> \delta n$. $n_0 >> \delta p$, is low-level injection.

(2) $n_0 = N_d = 10^{17}/cm^3$. $p_0 = \frac{n_i^2}{N_d} = 10^3/cm^3$

$$\begin{cases} n = n_i \cdot \exp\left(\frac{E_{Fn} - E_i}{kT}\right) \\ n = n_0 + \delta n \end{cases}$$

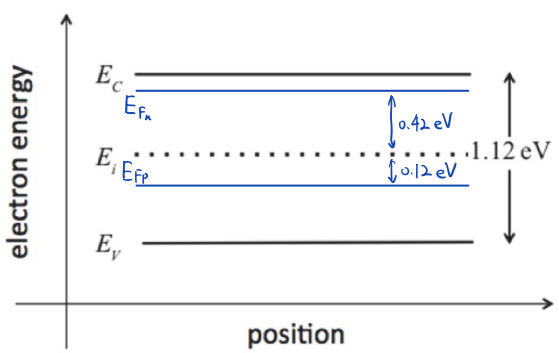
$\Rightarrow E_{Fn} = E_i + kT \cdot \ln \frac{n_0 + \delta n}{n_i}$

$E_{Fn} = E_i + 0.42\,eV$

$$\begin{cases} p = n_i \cdot \exp\left(\frac{E_i - E_{Fp}}{kT}\right) \\ p = p_0 + \delta p \end{cases}$$

$\Rightarrow E_{Fp} = E_i - kT \cdot \ln \frac{p_0 + \delta p}{n_i}$

$E_{Fn} = E_i - 0.12\,eV$



(3) At thermal equilibrium.

recombination rate $r = B \cdot n_0 \cdot p_0$

Carrier lifetime $\tau = \frac{1}{B \cdot n_0} = 10^{-6}\,s$

$\Rightarrow r = 10^9 \cdot cm^{-3} \cdot s^{-1}$

(4) generation rate : $g = Bn_i \cdot p_0 = 10^9 \cdot cm^{-3} \cdot s^{-1}$

For low level injection, $g_{net} \cong -B p_0 \cdot \delta n = -10^{18}\,cm^{-3} \cdot s^{-1}$

$g_{net} = g - r$

$\Rightarrow$ recombination rate at $\delta n = \delta p = 10^{12}\ cm^{-3}$. $r \approx 10^{18}\,cm^{-3} \cdot s^{-1}$

2. SRH recombination rate:

$$r_{net} = \frac{pn - n_i^2}{\tau_p \cdot [n + n_i \cdot \exp(\frac{E_T - E_i}{kT})] + \tau_n [p + n_i \cdot \exp(\frac{E_i - E_T}{kT})]}$$

while $E_T \ll E_i$, $n_i \exp(\frac{E_T - E_i}{kT}) \longrightarrow 0$, $n_i \cdot \exp(\frac{E_i - E_T}{kT}) \longrightarrow +\infty$

$r_{net} \longrightarrow 0$

while $E_T \ll E_i$, $n_i \exp(\frac{E_T - E_i}{kT}) \longrightarrow +\infty$, $n_i \cdot \exp(\frac{E_i - E_T}{kT}) \longrightarrow 0$

$r_{net} \longrightarrow 0$

$$\frac{\partial r_{net}}{\partial E_T} = \frac{\frac{pn - n_i^2}{kT} \cdot [-\tau_p n_i \cdot \exp(\frac{E_T - E_i}{kT}) + \tau_n n_i \cdot \exp(\frac{E_i - E_T}{kT})]}{[\tau_p \cdot [n + n_i \cdot \exp(\frac{E_T - E_i}{kT})] + \tau_n [p + n_i \cdot \exp(\frac{E_i - E_T}{kT})]]^2}$$

Assume $\tau_p = \tau_n$. while $E_T < E_i$. $\frac{\partial r_{net}}{\partial t} > 0$, while $E_T < E_i$. $\frac{\partial r_{net}}{\partial t} > 0$

$\Rightarrow r_{net}$ maximized at $E_T = E_i$

3. (1) $n = N_c \exp(-\frac{E_c - E_F}{kT})$ $\quad \frac{N_c}{N_s} = 0.28 < 1.$ $\quad N_s = \frac{4\pi}{h^3}(2m_n^* kT)^{\frac{3}{2}} \int_0^{+\infty} \sqrt{\eta} \cdot \frac{1}{1+\exp(\eta-\eta_f)} d\eta$

$\Rightarrow E_F = kT \ln \frac{n}{N_c} + E_c$

$N_c = 2(\frac{2\pi m_n^* kT}{h^2})^{\frac{3}{2}}$, For silicon at room temperature, $T = 300\,k$, $m_n^* = 1.08 m_0 = 9.84 \times 10^{-31}\,kg$.

$\Rightarrow N_c = 2.815 \times 10^{19}\,cm^{-3}$

① at $x = 0$, $N_s = 10^{20}\,cm^{-3}$

$n \cong N_s = 10^{20}\,cm^{-3}$

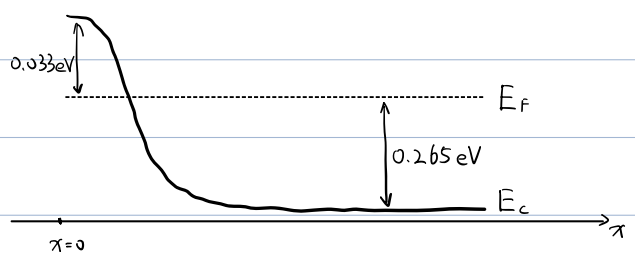
$E_F = E_c + 0.033\,eV$

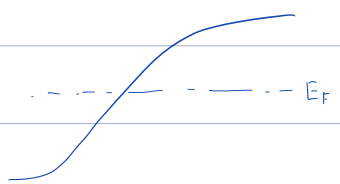
② at $x = \infty$, $N_d = 10^{15}\,cm^{-3}$

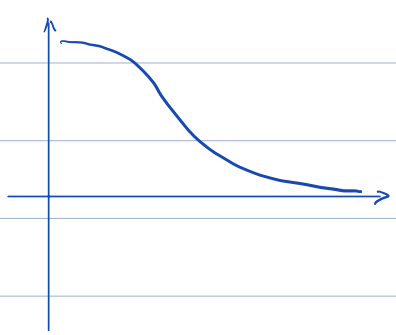
$n \cong N_d = 10^{15}\,cm^{-3}$

$E_F = E_c - 0.265\,eV$

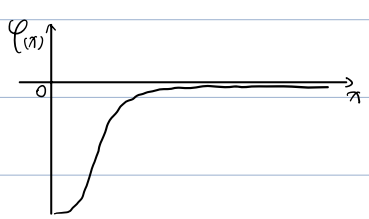
(2)







(3) Take $\varphi(\infty) = 0$



(4) Direction of electric field : $-x$ directions. $\quad \mathcal{E}(x) = -\frac{d\varphi(x)}{dx}$

